JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Differential Equations (2005–2014)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
21	109 M	63 Min	Classic Paper Pattern (2

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	21	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q16 • ID: JAM_2014_Q16) MCQ +2 M Neg: −0.67

Q.16 The value of $\alpha \in \mathbb{R}$ for which the curves $x^2 + \alpha y^2 = 1$ and $y = x^2$ intersect orthogonally is

(A) -2 (B) $\frac{-1}{2}$ (C) $\frac{1}{2}$ (D) 2

Question 2 (JAM 2014 • Q15 • ID: JAM_2014_Q15) MCQ +2 M Neg: −0.67

Q.15 The equation of the curve passing through the point $\left(\frac{\pi}{2}, 1\right)$ and having slope $\frac{\sin(x)}{x^2} - \frac{2y}{x}$ at each point (x, y) with $x \neq 0$ is

(A) $-x^2 y + \cos(x) = \frac{-\pi^2}{4}$ (B) $x^2 y + \cos(x) = \frac{\pi^2}{4}$

(C) $x^2 y - \sin(x) = \frac{\pi^2}{4} - 1$ (D) $x^2 y + \sin(x) = \frac{\pi^2}{4} + 1$

Question 3 (JAM 2014 • Q14 • ID: JAM_2014_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation

$$x^2 y''(x) - 2xy'(x) - 4y(x) = 0 \text{ for } x \in [1, 10].$$

Consider the Wronskian $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. If $W(1) = 1$, then $W(3) - W(2)$ equals

(A) 1 (B) 2 (C) 3 (D) 5

Question 4 (JAM 2014 • Q4 • ID: JAM_2014_Q4) MCQ +1 M Neg: −0.33

Q.4 For $a, b, c \in \mathbb{R}$, if the differential equation

$$\left(ax^2 + bxy + y^2\right)dx + \left(2x^2 + cxy + y^2\right)dy = 0$$

is exact, then

(A) $b = 2, c = 2a$ (B) $b = 4, c = 2$ (C) $b = 2, c = 4$ (D) $b = 2, a = 2c$

Question 5 (JAM 2012 • Q6 • ID: JAM_2012_Q6) MCQ +6 M Neg: −2.0

Q.6 An integrating factor for the differential equation $\left(2xy+3x^2y+6y^3\right)dx+\left(x^2+6y^2\right)dy=0$ is

(A) x^3 (B) y^3 (C) e^{3x} (D) e^{3y}

Question 6 (JAM 2012 • Q5 • ID: JAM_2012_Q5) MCQ +6 M Neg: −2.0

Q.5 The differential equation $\left(1+x^2y^3+\alpha x^2y^2\right)dx+\left(2+x^3y^2+x^3y\right)dy=0$ is exact if α equals

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Question 7 (JAM 2011 • Q6 • ID: JAM_2011_Q6) MCQ +6 M Neg: −2.0

Q.6 If y^a is an integrating factor of the differential equation $2xy\,dx - (3x^2-y^2)\,dy = 0$, then the value of a is

(A) −4 (B) 4 (C) −1 (D) 1

Question 8 (JAM 2011 • Q5 • ID: JAM_2011_Q5) MCQ +6 M Neg: −2.0

Q.5 The solution $y(x)$ of the differential equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$ satisfying the conditions $y(0)=4,\ \frac{dy}{dx}(0)=8$ is

(A) $4e^{2x}$ (B) $(16x+4)e^{-2x}$ (C) $4e^{-2x}+16x$ (D) $4e^{-2x}+16xe^{2x}$

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Question 9 (JAM 2010 • Q10 • ID: JAM_2010_Q10) MCQ +6 M Neg: −2.0

Q.10 Consider the differential equation $(x+y+1)\,dx+(2x+2y+1)\,dy=0$. Which of the following statements is true?

(A) The differential equation is linear
(B) The differential equation is exact
(C) e^{x+y} is an integrating factor of the differential equation
(D) A suitable substitution transforms the differentiable equation to the variables separable form

Question 10 (JAM 2010 • Q9 • ID: JAM_2010_Q9) MCQ +6 M Neg: −2.0

Q.9 Consider the differential equation $\frac{dy}{dx}=ay-by^2$, where $a,b>0$ and $y(0)=y_0$. As $x\to+\infty$, the solution $y(x)$ tends to

(A) 0 (B) $\frac{a}{b}$ (C) $\frac{b}{a}$ (D) y_0

Question 11 (JAM 2009 • Q14 • ID: JAM_2009_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $f,g:[-1,1]\to\mathbb{R}$, $f(x)=x^3$, $g(x)=x^2|x|$. Then

(A) f and g are linearly independent on $[-1,1]$
(B) f and g are linearly dependent on $[-1,1]$
(C) $f(x)g'(x)-f'(x)g(x)$ is NOT identically zero on $[-1,1]$
(D) there exist continuous functions $p(x)$ and $q(x)$ such that f and g satisfy $y''+py'+qy=0$ on $[-1,1]$

Question 12 (JAM 2009 • Q12 • ID: JAM_2009_Q12) MCQ +6 M Neg: −2.0

Q.12 Consider the differential equation $2\cos(y^2)\,dx-xy\sin(y^2)\,dy=0$. Then

(A) e^x is an integrating factor (B) e^{-x} is an integrating factor
(C) $3x$ is an integrating factor (D) x^3 is an integrating factor

Question 13 (JAM 2008 • Q14 • ID: JAM_2008_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $y_1(x)$ and $y_2(x)$ be twice differentiable functions on an interval I satisfying the differential equations $\frac{dy_1}{dx} - y_1 - y_2 = e^x$ and $2\frac{dy_1}{dx} + \frac{dy_2}{dx} - 6y_1 = 0$. Then $y_1(x)$ is

(A) $C_1 e^{-2x} + C_2 e^{3x} - \frac{1}{4}e^x$

(B) $C_1 e^{2x} + C_2 e^{-3x} + \frac{1}{4}e^x$

(C) $C_1 e^{2x} + C_2 e^{-3x} - \frac{1}{4}e^x$

(D) $C_1 e^{-2x} + C_2 e^{3x} + \frac{1}{4}e^x$

Question 14 (JAM 2008 • Q9 • ID: JAM_2008_Q9) MCQ +6 M Neg: −2.0

Q.9 Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I. Then $y_3(x) = a\,y_1(x) + b\,y_2(x)$ and $y_4(x) = c\,y_1(x) + d\,y_2(x)$ are linearly independent solutions of the given differential equation if

(A) $ad = bc$ (B) $ac = bd$ (C) $ad \neq bc$ (D) $ac \neq bd$

Question 15 (JAM 2008 • Q4 • ID: JAM_2008_Q4) MCQ +6 M Neg: −2.0

Q.4 The work done by the force $\vec{F} = 4y\,\hat{i} - 3xy\,\hat{j} + z^2\,\hat{k}$ in moving a particle over the circular path $x^2 + y^2 = 1,\ z = 0$ from $(1,0,0)$ to $(0,1,0)$ is

(A) $\pi + 1$ (B) $\pi - 1$ (C) $-\pi + 1$ (D) $-\pi - 1$

Question 16 (JAM 2007 • Q14 • ID: JAM_2007_Q14) MCQ +6 M Neg: −2.0

14. One of the integrating factors of the differential equation $(y^2 - 3xy)dx + (x^2 - xy)dy = 0$ is

(A) $1/(x^2y^2)$

(B) $1/(x^2y)$

(C) $1/(xy^2)$

(D) $1/(xy)$

Question 17 (JAM 2007 • Q2 • ID: JAM_2007_Q2) MCQ +6 M Neg: −2.0

2. If k is a constant such that $xy+k=e^{(x-1)^2/2}$ satisfies the differential equation $x\dfrac{dy}{dx}=(x^2-x-1)y+(x-1)$, then k is equal to

(A) 1
(B) 0
(C) −1
(D) −2

Question 18 (JAM 2006 • Q12 • ID: JAM_2006_Q12) MCQ +6 M Neg: −2.0

12. The differential equation representing the family of circles touching y-axis at the origin is

(A) linear and of first order
(B) linear and of second order
(C) nonlinear and of first order
(D) nonlinear and of second order

Question 19 (JAM 2006 • Q10 • ID: JAM_2006_Q10) MCQ +6 M Neg: −2.0

10. If $(c_1+c_2\ln x)/x$ is the general solution of the differential equation

$$x^2\frac{d^2y}{dx^2}+kx\frac{dy}{dx}+y=0, \qquad x>0,$$

then k equals

(A) 3
(B) −3
(C) 2
(D) −1

Question 20 (JAM 2005 • Q3 • ID: JAM_2005_Q3) MCQ +6 M Neg: −2.0

3. The general solution of $x^2 \dfrac{d^2y}{dx^2} - 5x\dfrac{dy}{dx} + 9y = 0$ is

(A) $(c_1 + c_2 x)e^{3x}$

(B) $(c_1 + c_2 \ln x)x^3$

(C) $(c_1 + c_2 x)x^3$

(D) $(c_1 + c_2 \ln x)e^{x^3}$

(Here c_1 and c_2 are arbitrary constants.)

Question 21 (JAM 2005 • Q2 • ID: JAM_2005_Q2) MCQ +6 M Neg: −2.0

2. An integrating factor of $x\dfrac{dy}{dx} + (3x+1)y = xe^{-2x}$ is

(A) $x\,e^{3x}$

(B) $3x\,e^{x}$

(C) $x\,e^{x}$

(D) $x^3 e^{x}$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	**D**
2	MCQ	+2	**B**
3	MCQ	+2	**D**
4	MCQ	+1	**B**
5	MCQ	+6	**C**
6	MCQ	+6	**B**
7	MCQ	+6	**A**
8	MCQ	+6	**B**
9	MCQ	+6	**D**
10	MCQ	+6	**B**
11	MCQ	+6	**A**

Q. No.	Type	Marks	Official Key
12	MCQ	+6	**D**
13	MCQ	+6	**C**
14	MCQ	+6	**C**
15	MCQ	+6	**D**
16	MCQ	+6	**B**
17	MCQ	+6	**A**
18	MCQ	+6	**C**
19	MCQ	+6	**A**
20	MCQ	+6	**B**
21	MCQ	+6	**A**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.